Cold Attack, Round 2
Borrowed-Edge Countermodels and the Anchor Residue

The mixed quadrant of the $\forall PO$-free fragment of $\mathcal{ALCI}_{\mathrm{RCC5}}$

Status:	Independent technical attack, not a referee report
Input packet:	`attack_mixed_quadrant_r2.zip`
Report date:	2026-08-28
Evidence:	Exact finite models, symbolic arguments, and executable checks

Principal result. Both open halves of Target D are false for the current borrowed-edge discipline. There is a refined-key blocked node whose gate-mate has no witness avoiding disjointness (D2), and there is a separate exact-key construction in which every fallback pair is PO but two groups form a directed cycle (D1). Target E splits: finite keyed node generation terminates, but the current gate-mate borrowing rule is not anchor-compatible and need not extend to a valid kernel frame.

Abstract

The round-2 packet reports substantial progress since the first attack: finite cross-kernel fusion is now Lean-formalized, and the down-spectrum gate proposed in round 1 has been integrated. The remaining construction borrows a gate-mate's witness for a blocked node. This report refutes both safety claims needed by that discipline. First, a seven-point mixed set model gives an actual refined-key blocked node for which the gate-mate's unique PP witness is DR from the borrower; the declared order stays acyclic but violates `ltNotDj`. Second, a 14-point set model gives two exact-key fallback groups whose borrowed pairs are all PO and individually safe, yet model-PP return paths make the declared order cyclic.

For Target E, the combinatorial node closure does terminate after at most one addition per key-demand pair. That finite external fixed point is not enough: a symmetric two-component model shows that gate-mate borrowing can declare two anchors disjoint although a selected kernel point lies below both, so no compatible kernel placement exists. The report also audits the supplied probes. In particular, `wp131` still uses the old type-only gate, aliases children across roles, and confounds agreement with repeat-free selection; `wp132`'s fresh-point evaluator is not a sound satisfiability test. Candidate repairs are isolated, but no complete extraction or decidability theorem is claimed.

Contents

1 Executive verdict and evidence ledger

The input packet distinguishes two borrowed-edge obligations and one anchor-closure question. The cold attack gives the following answers.

Question	Finding	Status
D1: global acyclicity	False for the local agreement-first/fallback discipline. Two individually safe PO fallback groups can alternate into a cycle.	Exact finite set-model counterexample [**X**]; a globally coordinated assignment remains possible.
D2: safe existing witness	False. At the exact down-spectrum key, the gate-mate's only relevant witness can be DR from the blocked node.	Exact mixed counterexample [**X**]; independent exhaustive core search.
E: finite closure	Pure keyed node generation terminates with the advertised finite bound.	Combinatorial lemma [**P**]/[**X**], essentially the packet's abstract gate theorem.
E: valid placement	The current gate-mate borrowing policy can produce a finite external frame that has no compatible kernel extension.	Exact finite counterexample [**X**]; general anchor-compatible completion remains open [**O**].
Probe evidence	Several headline measurements do not test the stated construction, and one wrapper masks failures.	Deterministic audit [**X**]; randomized counts remain measurements only [**R**].

1.1 Evidence levels

The tags used throughout are deliberately strict.

- [**P**] is a theorem reported by the packet as Lean-certified. The archive does not contain the Lean source, theorem statements, build manifest, or compiler output, so it was not independently rebuilt here.
- [**X**] is an exact finite model or finite combinatorial argument accompanied by a deterministic executable checker. It is mathematical evidence, but not a Lean proof.
- [**S**] is a symbolic infinite family, with bounded executable regression checks.
- [**R**] is a randomized finite sweep. It can find counterexamples but cannot prove a universal statement.

- **[O]** is an open construction or candidate reduction.

The packet reports a roughly 42,700-line Lean development with no sorries or warnings and axioms `propext`, `Quot.sound`, and `Classical.choice`. Those claims are treated as premises, not as independently reviewed results.

1.2 Continuity with round 1

Round 1's finite-rectangle result is reported formalized end to end as `atom_seq_stabilizes`, `two_tower_rectangle_gen`, `finite_fusion_recurrent`, and `fused_kq_all`. The packet also reports that the type-only common-top counterexample was accepted and that the down-spectrum key and local cone-transfer lemmas were certified. This report does not re-derive those results. It attacks the new claims that are logically downstream from them.

2 Exact setting and the quantifiers at issue

Let ρ be an atomic RCC5 model on a finite node set V. Strong equality is used: $\rho(x,y) = \mathsf{EQ}$ exactly when $x = y$. The convention is

$$\rho(x,y) = \mathsf{PP} \quad\Longleftrightarrow\quad x \text{ is a proper part of } y.$$

The refined gate key is

$$q(v) = (\mathrm{mty}(v),\ \{\mathrm{mty}(x) : \rho(x,v) = \mathsf{PP}\}) .$$

The gate retains the first node of each key. If v is blocked and a is its retained gate-mate, then $q(a) = q(v)$. For a nonpersistent demand $\delta = \exists\mathsf{PP}.D \in \mathrm{mty}(v)$, the current construction chooses one gate-mate witness z with

$$\rho(a,z) = \mathsf{PP}, \qquad z \models D,$$

and declares a borrowed order edge $v < z$.

The declared order seed is

$$L = \{(x,y) \in V^2 : \rho(x,y) = \mathsf{PP}\} \cup B,$$

where B is the set of borrowed edges. The frame must be ordered-disjoint:

- the transitive closure L^+ is irreflexive;
- `disj` is symmetric, irreflexive, and downward closed in both coordinates;
- $x < y$ implies $\neg\,\mathrm{disj}(x,y)$.

The packet's certified dichotomy concerns one edge: if a chosen z lies below v, then a lies below v, where an agreeing witness exists. This yields a local choice with $\rho(z,v) \neq \mathsf{PP}$. It does not settle either of the following:

D1. Do several locally safe borrowed edges keep L^+ irreflexive?

D2. In the non-agreeing case, can z always be chosen with $\rho(v,z) \neq \mathsf{DR}$ as well?

These are independent. D2 is a pairwise compatibility condition. D1 is a global graph condition, even if every borrowed pair is PO and hence already satisfies D2.

3 Target D2: the safe existing witness need not exist

Result [X]: D2 is false at the exact refined key. The failure uses a genuinely mixed, $\forall PO$-free input, an actual blocking run, a nonpersistent demand, a unique gate-mate witness, and equal nonempty lower spectra.

3.1 The seven-point mixed model

Use ordinary nonempty-set RCC5 semantics on the following points. Atomic propositions not displayed are false.

Point	Region	True atoms
a_0	$\{0\}$	—
a	$\{0, 1\}$	Q
z	$\{0, 1, 2\}$	P
v_0	$\{3\}$	—
v	$\{3, 4\}$	Q
w	$\{3, 4, 5\}$	P
p	$\{1, 4\}$	R

Let

$$C_0 = (\exists \mathsf{DR}.Q) \sqcap (\exists \mathsf{PP}.P) \sqcap (\exists \mathsf{PO}.R).$$

It contains no universal PO formula and lies in the mixed existential quadrant. Both a and v satisfy C_0:

- they are each other's unique $\mathsf{DR}.Q$ witnesses;
- z is the unique $\mathsf{PP}.P$ witness of a, and w is the unique $\mathsf{PP}.P$ witness of v;
- p is a $\mathsf{PO}.R$ witness of both.

The component swap

$$a_0 \leftrightarrow v_0, \qquad a \leftrightarrow v, \qquad z \leftrightarrow w, \qquad p \leftrightarrow p$$

is a valuation-preserving automorphism. Therefore $\mathrm{mty}(a) = \mathrm{mty}(v)$ over $\mathrm{cl}(C_0)$. Moreover, the strict lower cones of a and v are $\{a_0\}$ and $\{v_0\}$, and a_0, v_0 have the same closure type. Hence

$$q(a) = q(v),$$

with equal lower spectra of cardinality one. The example does not rely on an empty spectrum.

3.2 The actual gate run

Start with the duplicate-free singleton seed $[a]$. Its three existential demands have unique witnesses, so the first extension adds

$$v \quad (\mathsf{DR}.Q), \qquad z \quad (\mathsf{PP}.P), \qquad p \quad (\mathsf{PO}.R).$$

The refined gate scans $[a, v, z, p]$, keeps a first, and blocks v because $q(a) = q(v)$. The $\mathsf{PP}.P$ demand is not kernel-routed: its persistence guard

$$\forall\mathsf{PP}.(\exists\mathsf{PP}.P)$$

is false at a because the maximal point z has no $\mathsf{PP}.P$ successor.

The complete gate-mate candidate set is the singleton

$$\{y : \rho(a, y) = \mathsf{PP} \text{ and } y \models P\} = \{z\}.$$

But the two components are disjoint, so

$$\rho(v, z) = \rho(z, v) = \mathsf{DR}.$$

Consequently

$$\{y : \rho(a, y) = \mathsf{PP},\ y \models P,\ \rho(y, v) \neq \mathsf{PP},\ \rho(v, y) \neq \mathsf{DR}\} = \varnothing.$$

Declaring $v < z$ remains acyclic in this small model, which isolates D2 from D1. However, inherited disjointness contains v disj z, so the frame clause $v < z \Rightarrow \neg\,\mathrm{disj}(v, z)$ fails immediately.

3.3 What the counterexample rules out

The witness set is a singleton. Greedy, maximal, group-aware, and any other selection rule restricted to the gate-mate's existing witnesses make the same choice. Reversing which twin occurs first fails symmetrically. Thus the counterexample refutes the universal D2 existence claim, not merely one implementation of selection.

It does *not* refute expanding v and using its own witness w, adding a fresh class gadget, refining the key with additional incidence information, or changing the global blocking discipline. Those alternatives need new bounds and certificate proofs. The packaged exhaustive probe also finds a four-point network-level core and checks that no smaller domain occurs within universes of size at most three; this bounded minimality claim is executable evidence, not an unbounded theorem.

4 Target D1: locally safe fallback edges can cycle globally

Result [X]: D1 is false as a local-to-global implication. There is an exact down-spectrum-gated phase-1 run with two fallback groups, no common model-agreeing witness in either group, and every fallback pair PO, yet the declared order contains a directed cycle.

4.1 A 14-point principal-ideal set model

Let

$$\tau = (P \sqcup \neg P) \sqcap (Q \sqcup \neg Q), \qquad D = \exists\mathsf{PP}.\tau, \qquad C_0 = D \sqcap \tau.$$

There are no universal modalities. The formula is therefore $\forall PO$-free.

Use four minimal points $b_{00}, b_{01}, b_{10}, b_{11}$ below both a_1, a_2, and the non-transitive generating order

$$a_1 < z_1 < v_2 < t_2, \qquad a_2 < z_2 < v_1 < t_1, \qquad a_2 < z_2 < v_{1b} < t_{1b}.$$

Every named point is interpreted as its principal ideal, giving the following explicit regions and no unintended inclusions.

Point	Region	**Point**	Region
b_{00}	$\{0\}$	b_{01}	$\{1\}$
b_{10}	$\{2\}$	b_{11}	$\{3\}$
a_1	$\{0,1,2,3,4\}$	a_2	$\{0,1,2,3,5\}$
z_1	$\{0,1,2,3,4,6\}$	z_2	$\{0,1,2,3,5,7\}$
v_2	$\{0,1,2,3,4,6,8\}$	v_1	$\{0,1,2,3,5,7,9\}$
v_{1b}	$\{0,1,2,3,5,7,12\}$	t_2	$\{0,1,2,3,4,6,8,10\}$
t_1	$\{0,1,2,3,5,7,9,11\}$	t_{1b}	$\{0,1,2,3,5,7,12,13\}$

The two exposed atoms assign four colors to nonmaximal points:

$$\begin{array}{c|l} (0,0) & b_{00}, a_1, v_1, v_{1b} \\ (0,1) & b_{01}, a_2, v_2 \\ (1,0) & b_{10}, z_1 \\ (1,1) & b_{11}, z_2. \end{array}$$

The maximal t points have the displayed corresponding colors but lack D, since they have no strict upper point. All nonmaximal points satisfy D.

Each of $a_1, a_2, v_1, v_{1b}, v_2$ has all four nonmaximal closure types somewhere in its strict lower cone. Therefore the exact refined keys satisfy

$$q(a_1) = q(v_1) = q(v_{1b}), \qquad q(a_2) = q(v_2).$$

4.2 Phase-1 generation and the cycle

Start with the duplicate-free seed

$$[b_{00}, b_{01}, b_{10}, b_{11}, a_1, a_2].$$

First-witness expansion adds z_1, z_2, then adds v_2, v_1. The latter two repeat the exact keys of a_2, a_1 and are blocked; the run closes at the ten-node external set

$$V_0 = \{b_{00}, b_{01}, b_{10}, b_{11}, a_1, a_2, z_1, z_2, v_1, v_2\}.$$

The persistence guard $\forall \mathsf{PP}.D$ fails at both gate-mates, so the demand is edge-served.

No source-model point lies above both a_1 and v_1, and none lies above both a_2 and v_2. Thus group-aware agreement is unavailable. The deterministic fallback uses the gate children z_1, z_2 and declares

$$v_1 \to z_1, \qquad v_2 \to z_2.$$

Both borrowed pairs are PO in the source model. They satisfy the full D2 pairwise condition: the target is neither below nor disjoint from its borrower. Nevertheless, model inclusion contains $z_1 \mathsf{PP} v_2$ and $z_2 \mathsf{PP} v_1$, giving

$$v_1 \xrightarrow{B} z_1 \xrightarrow{\mathsf{PP}} v_2 \xrightarrow{B} z_2 \xrightarrow{\mathsf{PP}} v_1.$$

Hence L^+ is reflexive on v_1, z_1, v_2, z_2.

Adding v_{1b} gives two borrowers sharing z_1 and a second cycle through the same target. One target per group therefore does not remove cross-group cycles.

4.3 The exact global residue

Associate one vertex with each borrowed-target group. Draw a dependency

$$i \longrightarrow j$$

when target z_i is model-PP-below, or equal to, some borrower in group j. A declared cycle induces a cycle in this finite dependency graph; conversely, a dependency cycle expands to a cycle alternating borrowed edges and model-PP paths. Pairwise safety removes only self-loops. It says nothing about cycles of length at least two.

This counterexample refutes the current deterministic fallback and the inference from per-edge safety to global irreflexivity. It does not prove that every globally coordinated witness assignment cycles: later alternative witnesses exist in this model. A global rank, acyclic dependency selection, feedback-edge treatment, or finite CSP remains a possible repair.

5 Target E: finite closure and anchor placement

Result: E has two answers. Pure down-spectrum-gated node closure terminates with a finite bound. The current gate-mate borrowing policy nevertheless can produce a locally valid external fixed point that has no compatible kernel placement. General anchor-compatible completion remains open.

5.1 Positive combinatorial lemma: node closure terminates

Let Q be the finite key enumeration and Δ the relevant existential subformulas. If the gate consumes each pair $(q, \delta) \in Q \times \Delta$ at most once, then from any finite postselected anchor seed S_0 it adds at most one child for each such pair. Therefore

$$|E^*| \leq |S_0| + |Q|\,|\Delta| \leq |S_0| + N2^N\,|\operatorname{cl}(C_0)|.$$

The same product bounds the number of productive rounds. Once E^* is fixed, closing the order and disjointness relations on that finite carrier is noniterative with respect to point generation.

This is a set-theoretic termination statement. It is essentially the packet's reported abstract gate theorem. It does not say that the chosen edges and kernel incidences form a valid `MultiTier`.

5.2 Counterexample to the current anchor borrowing policy

Take two symmetric components $i \in \{0, 1\}$ with

$$p_{i0} < p_{i1} < p_{i2} < v_i, \qquad p_{i2} < u_i, \qquad v_i \;\mathsf{PO}\; u_i,$$

and declare every cross-component pair DR. Let X hold exactly at v_0, v_1, Y exactly at u_0, u_1, and Z at all six p points. Put

$$\begin{aligned} T &= \exists \mathsf{PP}.Z \sqcap \exists \mathsf{PP}.(X \sqcap \exists \mathsf{DR}.Y) \sqcap \exists \mathsf{PP}.Y, \\ C_0 &= T \sqcap \exists \mathsf{DR}.T. \end{aligned}$$

There are no universal modalities. The first two p points in each component satisfy C_0 and have equal full model type. Thus $p_{i0} < p_{i1}$ is an explicit recurrence segment, and the demand $\exists\mathsf{PP}.Z$ is served along the same-type p chain rather than being incidental. A component-swap automorphism gives

$$q(v_0) = q(v_1).$$

The $\mathsf{DR}.Y$ witnesses are uniquely crossed:

$$v_0 \xrightarrow{\mathsf{DR}} u_1, \qquad v_1 \xrightarrow{\mathsf{DR}} u_0.$$

Kernel i needs both v_i and u_i as PP anchors above its selected phase. If v_0 is the gate representative, blocked v_1 borrows v_0's witness u_1, so the external construction declares $v_1\mathsf{DR}u_1$. That declaration is locally harmless on the anchor-only carrier, where there are no order edges. It cannot coexist with $p_{11} < v_1$ and $p_{11} < u_1$, because

$$\mathsf{DR} \notin \mathrm{comp}(\mathsf{PPI}, \mathsf{PP}).$$

Equivalently, downward closure of disjointness would make a point below both disjoint from one of its strict uppers. If v_1 is retained first, the symmetric failure occurs in component 0.

Thus finite key enumeration bounds how many expansion tokens are consumed, but the key does not encode which kernel cone an anchor belongs to. The current policy can reach a finite external frame and still fail `hkex` placement.

The example does not refute all bounded completions. An existing-witness-first policy would choose u_0 for v_1 in this small model and avoid the clash, but no global bound or preservation theorem is known for that policy.

5.3 Why the old A–B family still matters

Let

$$A_n = \{0, \dots, n\}, \qquad B_m = \{m + 1\}.$$

For

$$H = (\exists\mathsf{PP}.\top) \sqcap (\exists\mathsf{DR}.\top), \qquad C_0 = H \sqcap \forall\mathsf{PP}.H,$$

all B_m have the same closure type and empty down-spectrum. Yet

$$\rho(A_n, B_m) = \begin{cases} \mathsf{DR}, & n \le m, \\ \mathsf{PPI}, & n \ge m + 1. \end{cases}$$

For any fixed finite source-model anchor set, choose n beyond every index used in it; then every anchor is a strict part of A_n, and none serves $\exists\mathsf{DR}.\top$. Thus a finite key space does not bound stabilization thresholds or produce a finite basis of source-model anchors [**S**].

This family does not by itself refute a fresh abstract anchor completion. Such a completion intentionally departs from source rows and must instead prove the anchor-compatibility lemma exposed above.

6 Audit of the supplied probes

The supplied probes are useful exploratory artifacts, but several headline readings are not measurements of the stated R2 construction. The original files are preserved unchanged in the package; corrected and independent checks are kept separately.

6.1 `wp131`: four material defects

1. **Wrong gate.** The target uses $q = (\text{mty}, \text{lower spectrum})$, but `gated()` and gate-mate lookup in the supplied script use mty alone. The published 1,040-instance, 132-fallback result is therefore a type-only measurement, not a Target-D measurement.
2. **The child cache omits the role.** It is keyed by (a, body) rather than $(a, \text{role}, \text{body})$. If the same body occurs under two modalities, a PP borrower can be redirected to another role's child. In the deterministic supplied sweep, 18 of 1,772 borrowed targets were not PP witnesses of the blocker. Their relations from blocker to target were

$$\mathsf{DR} : 8, \qquad \mathsf{PO} : 3, \qquad \mathsf{EQ} : 6, \qquad \mathsf{PPI} : 1.$$

   Adding the role changes the type-only headline counts: greedy cyclic instances fall from 522 to 515, and repeat-free survivors from 17 to 8.
3. **Part G tests individuals, not groups.** The group-aware rule requires one witness above all borrowers in a group. Part G instead asks whether some borrower lacks an individual agreeing witness. The supplied predicate reports 132 fallback instances; the group predicate gives 135. With the actual refined key and relation-sensitive child cache, the corresponding counts are 164 and 167.
4. **Part F confounds two interventions.** Its "agree" column calls `repeat_free=True` together with `mode="agree"`. Pure group-aware agreement has one cycle in the supplied type-only sweep; only the combination has zero. The corrected refined-key finite sweep happens to have zero for both, but that remains randomized evidence and cannot imply D1, which the exact model in Section 4 refutes.

The non-vacuity pass is also forced: sampling is conditioned on at least one borrowed edge, then Part N accepts a class averaging at least 0.5 borrowed edges. Every retained class averages at least one. The reported retention rates remain informative.

6.2 `wp132`: a regression, not a placement test

The function `fresh_sat` returns true for every non-DR universal and false for every existential. It therefore accepts

$$\forall\mathsf{EQ}.(P \sqcap \neg P)$$

and rejects $\exists\mathsf{EQ}.P$ even when the fresh assignment makes P true. It is neither sound nor complete as a satisfiability evaluator.

The existing-point route checks only that a point is DR from a lower-cone union and satisfies the body in the old model. It does not construct the shared top or check truth and composition after extension. The fresh route likewise performs no RCC5 extension check. Of the 7,525 reported refined-key group-demand evaluations, 6,038 (80.2 percent) are accepted only by this fresh abstraction and 1,487 use an existing point. These are correlated evaluations from 5,771 accepted generated models.

Fairly interpreted, the generated bodies are restricted to disjunctions of $\forall\mathsf{DR}$ atoms, so the displayed EQ and existential errors do not directly invalidate that narrow sweep. The probe is a purpose-built regression of local lower-cone agreement on its generated family. It is not evidence for a realizable copied gadget, full serviceability, or Target E.

6.3 The wrapper masks failures

The supplied `run_probes.sh` executes

```
python3 "$f" || echo "(non-zero exit ...)".
```

This converts exceptions and nonzero verdicts into shell success, defeating `set -e`. A zero wrapper exit status is not a correctness verdict.

7 Bounded repair directions and their exact obligations

7.1 No stronger existing-witness selector repairs D2

Section 3's candidate set is a singleton. Any repair confined to choosing another gate-mate witness is impossible. At least one of the following must change: whether v is blocked, which points are available, the key/grouping discipline, or the use of the source witness as the borrowed target.

Refusing to block DR-related same-key nodes is not automatically bounded. There can be arbitrarily many automorphic, pairwise-DR leaves with private P tops. A unary type/down-spectrum refinement does not see their multiplicity or pairwise incidence.

7.2 A global dependency discipline for D1

For a fixed finite external set, build the borrowed-target dependency graph from Section 4. A globally coordinated selector can reject assignments with cyclic dependency graphs, introduce a well-founded rank, or compute a feedback-edge set. Acyclic components need no new D1 proof; cyclic strongly connected components are exactly where fresh gadgets, unrolling, or kernelization are required.

This characterization is finite and checkable, but existence of an acyclic assignment with the required labels and all frame obligations remains open.

7.3 Fresh class-demand gadgets

A plausible repair replaces an unsafe model witness by a fresh top $t_{q,\delta}$ for each refined key q and active PP demand δ, placing every borrower in that class below the fresh top. The packet's reported `dkey_union_serves` lemma is exactly the local tool that the refuted type-only common top lacked. If $K = N2^N$ and there are $p \leq |\operatorname{cl}(C_0)|$ active PP demands, the first layer uses at most Kp fresh roots.

This is a bounded program, not a completed construction. Each fresh gadget has its own demands; recursive closure can regenerate the same problem. It must also preserve already stabilized kernel rows and package genuine recurrent suffixes as `KernelData`. Private clones without this accounting do not prove termination.

7.4 Conditional witness-incidence and kernel-profile keys

If a finite witness pool W is fixed in advance, refine the key to

$$q_W(v) = (q(v), \{w \in W : \rho(v, w) = \mathsf{PP}\}) .$$

Then $q_W(a) = q_W(v)$ and $\rho(a, z) = \mathsf{PP}$ for $z \in W$ imply $\rho(v, z) = \mathsf{PP}$ literally. Every borrowed order edge agrees with the source model, so D1 and D2 disappear. The number of keys is at most

$$N2^N\, 2^{|W|}.$$

The condition is circular: demand-closing W can add witnesses, change the key, and create new representatives. Target E is precisely the missing finite-pool theorem.

After kernel segments are selected, an alternative is to add each anchor's finite kernel-incidence profile. Recording one RCC5 atom per kernel costs at most a $5^{|\kappa|}$ factor and separates the Target-E counterexample. All abstract gate, borrowed-edge, and frame lemmas would need to be reproved for that profile.

Weakening `ltNotDj`, simply deleting conflicting DR seeds, sending arbitrary nonrecurrent demands to kernels, or adding unclosed private clones does not address the actual obligations.

8 Integrated status and recommended next steps

The round-2 packet made real progress: the finite cross-kernel fusion theorem and the down-spectrum cone-transfer mechanism are reported formalized. The present attack shows that the next local selection step is not sound and that finite key enumeration does not by itself solve semantic anchor placement.

Item	Defensible status after this attack
Target B / `kq_all`	Reported Lean-certified by R2; not rebuilt from this archive.
Down-spectrum local cone transfer	Reported Lean-certified and not refuted here.
Target D1	Current local fallback proof route refuted by a PO-only alternating cycle.
Target D2	Universal existing-witness selection refuted by a unique-witness mixed counterexample.
Target E node termination	Finite combinatorial bound available.
Target E placement / `hkex`	Current borrowing policy refuted; general anchor-compatible completion open.
Final packaging and extraction	Still open; the packet itself warns that several "bookkeeping" steps remain.

The most efficient research order is now:

1. formalize the D1 and D2 finite countermodels as regression theorems;
2. repair `wp131`, separate selection interventions, and make the wrapper propagate failures;
3. state an anchor-compatible gate theorem with explicit kernel-incidence data;
4. choose between a globally acyclic witness assignment, fresh class-demand gadgets, or a fixed-pool/profile construction, and prove its finite closure;
5. only then rebuild `KernelData`, reindex, and assemble the final `MergedExtractionAt` object.

No complete extraction theorem for the mixed quadrant, and no decidability theorem for full $\mathcal{ALCI}_{\mathrm{RCC5}}$, is claimed.

A Artifacts, reproducibility, and scope

The accompanying ZIP preserves the supplied R2 packet unchanged and adds independent checks:

- `target_d1_multigroup_cycle.py`: exact 14-point D1 model, phase-1 seed expansion, refined keys, group-aware fallback, and alternating cycles;
- `target_d2_mixed_counterexample.py`: direct mixed D2 model and fixed-point gate run;
- `target_d2_exhaustive_countermodel.py`: bounded core search plus formula-realized and nonempty-spectrum strengthenings;
- `target_e_anchor_compatibility.py`: symmetric kernel-anchor clash;
- `target_e_finite_key_anchor_counterexample.py`: symbolic A_n/B_m finite-anchor calculation;
- `wp131_refined_key_audit.py`: the packet sweep with the refined key, relation-sensitive child cache, and group-level fallback count;
- `probe_audit_r2.py`: deterministic reproduction of the omitted predicates and `wp132` evaluator checks.

From the package root, run

`./run_new_probes.sh`.

Every packaged file except the manifest itself is covered by `MANIFEST.sha256`. The exact finite countermodels are independent of the randomized seed used by the exploratory sweeps.

A.1 Scope of the countermodels

All exact countermodels use ordinary nonempty finite sets, so RCC5 converse and weak composition closure follow from concrete semantics and are also checked exhaustively on the displayed finite domains. The probes separately verify strong equality, the source ordered-disjoint axioms, formula satisfaction, closure-type equality, exact lower spectra, persistence guards, gate order, complete candidate sets, and the claimed final failure.

The D1 model refutes the displayed deterministic fallback and the universal implication from local pair safety to global acyclicity. It intentionally does not claim that every possible simultaneous witness assignment fails. The D2 model is stronger with respect to selection: the relevant candidate set is a singleton, so no selection among existing gate-mate witnesses can satisfy D2. The Target-E model refutes the present borrowing policy, while leaving open a different bounded anchor-compatible construction.